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GAMING DEVICE WITH A USB INTERFACE FOR DELIVERING POWER AND DATA TO VIDEO DISPLAY MODULES

Abstract

A gaming device uses a USB interface for power delivery and data transmission. In non-limiting examples, the USB interface may be a universal serial bus (USB) C-type cable communicatively and electrically connecting multiple electronic gaming devices and/or auxiliary devices, such as monitors, thereby reducing the number of cables for power and data delivery between machines and/or devices.

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Background/Summary

RELATED APPLICATION

[0001] An Application Data Sheet is filed concurrently with this specification as part of the present application. Each application that the present application claims benefit of or priority to as identified in the concurrently filed Application Data Sheet is incorporated by reference herein in their entireties and for all purposes.

BACKGROUND

[0002] Electronic gaming machines ("EGMs") or gaming devices provide a variety of wagering games such as slot games, video poker games, video blackjack games, roulette games, video bingo games, keno games and other types of games that are frequently offered at casinos and other locations. Play on EGMs typically involves a player establishing a credit balance by inputting money, or another form of monetary credit, and placing a monetary wager (from the credit balance) on one or more outcomes of an instance (or single play) of a primary or base game. In some cases, a player may qualify for a special mode of the base game, a secondary game, or a bonus round of the base game by attaining a certain winning combination or triggering event in, or related to, the base game, or after the player is randomly awarded the special mode, secondary game, or bonus round. In the special mode, secondary game, or bonus round, the player is given an opportunity to win extra game credits, game tokens or other forms of payout. In the case of "game credits" that are awarded during play, the game credits are typically added to a credit meter total on the EGM and can be provided to the player upon completion of a gaming session or when the player wants to "cash out."

[0003] "Slot" type games are often displayed to the player in the form of various symbols arrayed in a row-by-column grid or matrix. Specific matching combinations of symbols along predetermined paths (or paylines) through the matrix indicate the outcome of the game. The display typically highlights winning combinations/outcomes for identification by the player. Matching combinations and their corresponding awards are usually shown in a "pay-table" which is available to the player for reference. Often, the player may vary his/her wager to include differing numbers of paylines and/or the amount bet on each line. By varying the wager, the player may sometimes alter the frequency or number of winning combinations, frequency or number of secondary games, and/or the amount awarded.

[0004] Typical games use a random number generator (RNG) to randomly determine the outcome of each game. The game is designed to return a certain percentage of the amount wagered back to the player over the course of many plays or instances of the game, which is generally referred to as return to player (RTP). The RTP and randomness of the RNG ensure the fairness of the games and are highly regulated. Upon initiation of play, the RNG randomly determines a game outcome and symbols are then selected which correspond to that outcome. Notably, some games may include an element of skill on the part of the player and are therefore not entirely random.

SUMMARY

[0005] A gaming device may include a cabinet construction containing a host control module and a plurality of individual components (e.g., multiple monitors, speakers, lighting devices, cameras, microphones, etc.). In conventional gaming devices, each one of these components may be communicatively and electrically connected to the host control module via multiple separate cables that perform different functions. For example, a monitor having a resolution of at least 3840×2160 at 60 frames or more per second may be communicatively and electrically connected to the host control module via a multiple power cables and multiple separate data cables that transmit data (e.g., data cables transmitting video signals via DisplayPort, USB connections for touch communication, USB connections for edge lighting, USB connections for camera input, USB

connections for microphone input, etc.). Because each component may connect to the host control module via multiple cables and the gaming device may have multiple components, it may be cumbersome to assemble, repair, and maintain the gaming device. In other implementations, the monitor may a resolution greater or less than 3840×2160 at 60 frames or more per second (e.g., where the gaming device includes button decks and/or a main monitor is a laser-cut LCD for a custom aspect ratio and thus a custom resolution, etc.).

[0006] In the present disclosure, the gaming device may use a single USB-C cable in connection between a host control module and a first video display module and an additional single USB-C cable in connection between the first video display module and a second video display module for power delivery and data transmission. Each module may include a monitor and many peripheral components of the electronic gaming machine (EGM). The USB-C cable may implement a USB connector system that combines multiple protocols in a single cable to offload many peripheral component connections from the EGM into the module, which simplifies connections and enables a reduction in the number of cables to one USB-C cable connecting one module to another module. The use of a single cable may simplify connections thereby simplifying repair or maintenance and decreasing associated costs. Furthermore, the disclosed single USB-C cable may be capable of delivering 100 W, 140 W, 180 W, 240 W, or more of power, thereby enabling a single cable to deliver power to more components at higher power levels than conventional designs.

[0007] A gaming device is provided that may include a host control module having a host USB controller, a game controller communicatively connected to the host USB controller to transmit data, and a power source electrically connected to the host USB controller to deliver power. The gaming device may further include a first video display module having a first USB controller and a first video monitor communicatively and electrically connected to the first USB controller. The gaming device may optionally further include a second video display module having a second USB controller and a second video monitor communicatively and electrically connected to the second USB controller. The first USB controller may be configured to receive, via a first USB interface, first power and first data originating from the host USB controller and to transmit the first data and deliver at least some of the first power to the first video monitor. The second USB controller may be configured to receive, via a second single USB interface, second power and second data originating from the host USB controller and to transmit the second data and to deliver at least some of the second power to the second video monitor and the one or more second peripheral components. A Unified USB Data and Power Distribution System may include the host USB controller, the first USB controller, and the second USB controller. The Unified USB Data and Power Distribution System may include a plurality of processors and a plurality of memory devices. The one or more memory devices may store computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the processors to collectively deliver at least the first power to the first USB controller via the first USB interface responsive, at least in part, to the one or more processors receiving a request for the first power from the first USB controller via the first USB interface. The one or more memory devices may also store computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the processors to collectively allocate at least some of the first power to the first video monitor and at least some of the second power to the second video monitor responsive, at least in part, to the first USB controller receiving the first power originating from the host USB controller and the second USB controller receiving the second power originating from the host USB controller.

[0008] The host control module may be communicatively and electrically connected to the first video display module, and the first video display module may be communicatively and electrically connected to the second video display module, thereby connecting the host control module, the first video display module, and the second video display module in a serial combination.

[0009] The memory devices may store computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the processors to deliver an amount of power to the first video display module, the amount of power being at least the sum total of the first power and the second power. The memory devices may also store computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the processors to allocate at least the second power from the first USB controller to the second USB controller responsive, at least in part, to the first video display module receiving the amount of power.

[0010] The host control module may be communicatively and electrically connected directly to the first video display module, and the host control module may be further communicatively and electrically connected directly to the second video display module, thereby communicatively and electrically connecting the first video display module and the second video display module in a parallel combination with the host USB controller.

[0011] The memory devices may store computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the processors to deliver an amount of power to the first video display module, with the amount of power being at least the first power. The memory devices may store computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the processors to deliver an additional amount of power to the second video display module, the additional amount of power being at least the second power.

[0012] The first USB interface and/or the second USB interface may include a USB cable capable of delivering at least 240 W and carrying a data transfer speed of at least 40 Gbps.

[0013] The one or more first peripheral components may be communicatively and electrically connected to the first USB controller, and the first USB controller may be configured to transmit the first data and deliver at least some of the first power from the first USB controller to the one or more first peripheral components.

[0014] The one or more second peripheral components may be communicatively and electrically connected to the second USB controller, and the second USB controller may be configured to transmit the second data and deliver at least some of the second power to the one or more second peripheral components.

[0015] The first USB controller may include a first power delivery subsystem. The first power delivery subsystem may be configured to negotiate, via the first USB interface, with the host USB controller to request the first power.

[0016] The first video monitor and the one or more first peripheral components may be configured to collectively negotiate with the first power delivery subsystem to request the first power. The first power is at least a sum total of a plurality of first individual power levels corresponding with the first video monitor and the one or more first peripheral components.

[0017] The second USB controller may include a second power delivery subsystem. The second power delivery subsystem may be configured to negotiate, via the second USB interface, with the first USB controller to request the second power.

[0018] The second video monitor and the one or more second peripheral components may be configured to collectively negotiate with the second power delivery subsystem to request the second power. The second power may be at least a sum total of a plurality of second individual power levels corresponding with the second video monitor and the one or more second peripheral components.

[0019] The one or more memory devices may further store additional computer-executable instructions which, when executed by the one or more processors, further cause the one or more processors to compare a sum total of the first power and the second power to a maximum power level capacity of one or both of the first USB interface and the second USB interface. The one or more memory devices may further store additional computer-executable instructions which, when executed by the one or more processors, further cause the one or more processors to generate data associated with a remedial action responsive, at least in part, to the host USB controller determining that the sum total of the first power and the second power exceeds the maximum power level capacity.

[0020] The first USB controller may include a first video logic subsystem configured to generate an input signal responsive, at least in part, to the first video logic subsystem receiving the data associated with the remedial action from the processors. The first video display module may further include a display control subsystem configured to generate a video signal responsive, at least in part, to the display control subsystem receiving the input signal from the first video logic subsystem. The first video monitor may present a notification responsive, at least in part, to the first video monitor receiving the video signal from the display control subsystem. The notification may include a prompt to install an additional USB interface to connect the host USB controller with the first video display module or the second video display module.

[0021] The second USB controller may include a second video logic subsystem configured to generate an input signal responsive, at least in part, to the second video logic subsystem receiving the data associated with the remedial action from the processors. The second video display module may further include a display control subsystem configured to generate a video signal responsive, at least in part, to the display control subsystem receiving the input signal from the second video logic subsystem. The second video monitor may present a notification responsive, at least in part, to the second video monitor receiving the video signal from the display control subsystem. The notification may include a prompt to install an additional USB interface to connect the host USB controller with the first video display module or the second video display module.

[0022] The first USB controller may include a first power delivery subsystem that is configured to stop delivering at least some of the first power to one or more of the first peripheral components responsive, at least in part, to the first USB controller receiving the data associated with the remedial action from the host USB controller.

[0023] The second USB controller may include a second power delivery subsystem that is configured to stop delivering at least some of the second power to one or more of the second peripheral components responsive, at least in part, to the second USB controller receiving the data associated with the remedial action from the first USB controller.

[0024] The first video monitor and/or the second video monitor may have a resolution of at least 3840×2160 and that supports a frame rate of at least 60 frames per second.

[0025] At least one of the first peripheral components and the second peripheral components may include a touchscreen display, a lighting device, a microphone, a camera, a speaker, and/or a supplemental monitor.

[0026] The first peripheral components and/or the second peripheral components may include a DisplayPort connector supporting a monitor having a resolution of at least 3840×2160 at 60 frames per second, a USB-A connector for touch control data, a USB-A connector for lighting data, a USB-A connector for mono microphone data, a USB-A connector for a full HD at 30 frames per second camera, a USB-A connector for USB audio, a USB-C Power Delivery connector for an additional monitor, an HDMI connector, and/or a power output for the first video monitor or the second video monitor via a Molex/Amphenol header.

[0027] A method may be provided for configuring a gaming device. The method may include transmitting, using a first USB interface, first power and first data originating from a host USB controller to a first USB controller of a first video display module. The method may further include transmitting the first data and at least some of the first power from the first USB controller to a first video monitor of the first video display module. The method may further include transmitting, using a second USB interface, second power and second data originating from the host USB controller to a second USB controller of a second video display module. The method may further include transmitting the second data and at least some of the second power from the second USB controller to a second video monitor.

[0028] The method may further include transmitting the first data and at least some of the first power from the first USB controller to one or more first peripheral components.
[0029] The method may further include transmitting the second data and at least some of the second power from the second USB controller to one or more second peripheral components.
[0030] The method may further include negotiating, using a first power delivery subsystem via the first USB interface, with the host USB controller to request the first power.
[0031] The method may further include negotiating, using the first video monitor and the one or more first peripheral components collectively, with the first delivery power subsystem to request the first power. The first power is at least a sum total of a plurality of first individual power levels corresponding with the first video monitor and the one or more first peripheral components.
[0032] The method may further include negotiating, using a second power delivery subsystem of the second USB controller via the second USB interface, with the first USB controller to request the second power.
[0033] The method may further include negotiating, using the second video monitor and the one or more second peripheral components collectively, with the second power delivery subsystem to request the second power. The second power is at least a sum total of a plurality of second individual power levels corresponding with the second video monitor and the one or more second peripheral components.
[0034] The method may further include comparing, using the host USB controller, the amount of power to maximum power level capacity of the USB interface. The method may further include generating, using the host USB controller, data associated with a remedial action responsive, at least in part, to the host USB controller determining that the amount of power exceeds the maximum power level capacity.
[0035] The method may further include generating, using a first video logic subsystem of the first USB controller, an input signal responsive, at least in part, to the first video logic subsystem receiving the data associated with the remedial action from the host USB controller. The method may further include generating, using a display control subsystem of the first video display module, a video signal responsive, at least in part, to the display control subsystem receiving the input signal from the first video logic subsystem. The method may further include presenting, using the first video monitor, a notification responsive, at least in part, to the first video monitor receiving the video signal from the display control subsystem. The method may further include prompting, using the notification, an installation of an additional USB interface to connect the host USB controller with the first video display module or the second video display module.
[0036] The method may further include generating, using a second video logic subsystem of the second USB controller, an input signal responsive, at least in part, to the second video logic subsystem receiving the data associated with the remedial action from the first USB controller. The method may further include generating, using a display control subsystem of the second video display module, a video signal responsive, at least in part, to the display control subsystem receiving the input signal from the second video logic subsystem. The method may further include presenting, using the second video monitor, a notification responsive, at least in part, to the second video monitor receiving the video signal from the display control subsystem. The method may further include prompting, using the notification, an installation of an additional USB interface to connect the host USB controller with the first video display module or the second video display module.
[0037] The method may further include stopping, using a first power delivery subsystem of the first USB controller, delivery of at least some of the first power to the first video monitor or one or more of the first peripheral components responsive, at least in part, to the first USB controller receiving the data associated with the remedial action from the host USB controller.
[0038] The method may further include stopping, using a second power delivery subsystem of the second USB controller, delivery of power to the second video monitor or one or more of the second

peripheral components responsive, at least in part, to the second USB controller receiving the data associated with the remedial action from the first USB controller.

[0039] One or more non-transitory computer-readable media may store computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause a first power originating from a host USB controller of a host control module to be delivered, via a first USB interface, to a first USB controller of a first video display module responsive, at least in part, to the processor receiving from the first USB controller a request for the first power. The one or more non-transitory computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause at least some of the first power to be allocated from the first USB controller to a first video monitor of the first video display module responsive, at least in part, to the first USB controller receiving the first power originating from the host USB controller. The one or more non-transitory computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause a second power originating from the host USB controller of the host control module to be delivered, via a second USB interface, to a second video display module responsive, at least in part, to the processor receiving from the first USB controller a request for the first power. The one or more non-transitory computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause at least some of the second power to be allocated from the second USB controller to a second video monitor of the second video display module responsive, at least in part, to the second USB controller receiving the second power originating from the host USB controller.

[0040] The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause a first video monitor and one or more first peripheral components to negotiate with the first USB controller to collectively request the first power, and the first power being at least a sum total of a plurality of first individual power levels corresponding with the first video monitor and the one or more first peripheral components.

[0041] The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause a first power delivery subsystem of the first USB controller to negotiate, via the first USB interface, with the host USB controller to request at least the first power.

[0042] The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause a second video monitor and one or more second peripheral components to negotiate with a second USB controller to request the second power. The second power is at least a sum total of a plurality of second individual power levels corresponding with the second video monitor and the one or more second peripheral components.

[0043] The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause a second power delivery subsystem of the second USB controller to negotiate, via the second USB interface, with the first USB controller to request the second power.

[0044] The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause the host USB controller to compare a sum total of the first power and the second power to a maximum power level capacity of one or both the first USB interface and the second USB interface. The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause, responsive, at least in part, to the host USB controller determining that the sum total of the first power and the second power exceeds the maximum power level capacity, the host USB controller to generate data associated with a remedial action.

[0045] The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause, responsive, at least in part, to a first video logic subsystem of the first USB controller receiving data associated with the remedial action, the first video logic subsystem to generate an input signal. The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause, responsive, at least in part, to a display control subsystem of the first video display module receiving the input signal from the first video logic subsystem, the display control subsystem to generate a video signal. The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause, responsive, at least in part, to the first video monitor receiving the video signal from the display control subsystem, the first video monitor to present a notification. The notification may include a prompt to install an additional USB interface to connect the host USB controller with the first video display module or the second video display module.

[0046] The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause, responsive, at least in part, to a second video logic subsystem of the second USB controller receiving data associated with the remedial action, the second video logic subsystem to generate an input signal. The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause, responsive, at least in part, to a display control subsystem of the second video display module receiving the input signal from the second video logic subsystem, the display control subsystem to generate a video signal. The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause, responsive, at least in part, to a second video monitor receiving the video signal from the display control subsystem, the second video monitor to present a notification. The notification may include a prompt to install an additional USB interface to connect the host USB controller with the first video display module or the second video display module.

[0047] The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause, responsive, at least in part, to a first power delivery subsystem of the first USB controller receiving data associated with the remedial action from the host USB controller, the first power delivery subsystem to stop delivering at least some of the first power to one or more first peripheral components.

[0048] The one or more computer-readable media may store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to cause, responsive, at least in part, to a second power delivery subsystem of the second USB controller receiving data associated with the remedial action from the first USB controller, the second power delivery subsystem to stop delivering at least some of the second power to one or more of second peripheral components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0049] FIG. **1** is an exemplary diagram showing several EGMs networked with various gaming related servers.

[0050] FIG. **2**A is a block diagram showing various functional elements of an exemplary EGM.

[0051] FIG. **2**B depicts a casino gaming environment according to one example.

[0052] FIG. **2**C is a diagram that shows examples of components of a system for providing online gaming according to some aspects of the present disclosure.

[0053] FIG. **3** illustrates, in block diagram form, an implementation of a game processing architecture algorithm that implements a game processing pipeline for the play of a game in accordance with various implementations described herein.

[0054] FIG. **4** depicts, in block diagram form, an implementation of a gaming device having USB interfaces connecting a host control module, a first video display module, and a second video display module in a serial combination for power delivery and data transmission.

[0055] FIG. **5** depicts, in block diagram form, another implementation of the gaming device of FIG. **4**, illustrating USB interfaces connecting a first video display module and a second video display module in a parallel combination with a host control module for power delivery and data transmission.

[0056] FIG. **6** depicts, in block diagram form, yet another implementation of the gaming device of FIG. **4**, illustrating the gaming device having a single video display module.

[0057] FIG. **7** depicts a flow chart of an example method of configuring the gaming device of FIG. **4**, including using a first USB-C cable to negotiate for power from a host to a connected first video display module and providing suitable power to a second video display module from the first video display module when the second USB-C cable is connected via negotiation.

DETAILED DESCRIPTION

[0058] A gaming device uses a single USB-C cable in connection between a host control module and a first video display module and an additional single USB-C cable in connection between the first video display module and a second video display module for power delivery and data transmission. Each module may include a monitor and many peripheral components of the electronic gaming machine (EGM). The USB-C cable may implement a 24-pin USB connector system with a rotationally symmetric connector that combines multiple protocols in a single cable (including DisplayPort, peripheral component interconnect express (PCIe) or Thunderbolt 4 protocol). This enables offloading many peripheral components from the EGM into the module, as well as reducing the number of cables previously going into the module from multiple cables down to one USB-C cable.

[0059] The disclosed single USB-C cable provides advantages over conventional systems with modules which employ multiple ports for various protocols as well as dedicated cabling between each port. The use of a single cable simplifies connections and thereby simplifies manufacturing, repair or maintenance, and reduces component cost. Furthermore, the disclosed single USB-C cable is capable of delivering 100 W of power, 140 W of power, 180 W of power, 240 W of power, or more, enabling a single cable to deliver power to more components at higher power levels than conventional designs.

[0060] FIG. **1** illustrates several different models of EGMs which may be networked to various gaming related servers. Shown is a system **100** in a gaming environment including one or more server computers **102** (e.g., slot servers of a casino) that are in communication, via a communications network, with one or more gaming devices **104**A-**104**X (EGMs, slots, video poker, bingo machines, etc.) that can implement one or more aspects of the present disclosure. The gaming devices **104**A-**104**X may alternatively be portable and/or remote gaming devices such as, but not limited to, a smart phone, a tablet, a laptop, or a game console. Gaming devices **104**A-**104**X utilize specialized software and/or hardware to form non-generic, particular machines or apparatuses that comply with regulatory requirements regarding devices used for wagering or games of chance that provide monetary awards.

[0061] Communication between the gaming devices **104**A-**104**X and the server computers **102**, and among the gaming devices **104**A-**104**X, may be direct or indirect using one or more communication protocols. As an example, gaming devices **104**A-**104**X and the server computers

102 can communicate over one or more communication networks, such as over the Internet through a website maintained by a computer on a remote server or over an online data network including commercial online service providers, Internet service providers, private networks (e.g., local area networks and enterprise networks), and the like (e.g., wide area networks). The communication networks could allow gaming devices **104**A-**104**X to communicate with one another and/or the server computers **102** using a variety of communication-based technologies, such as radio frequency (RF) (e.g., wireless fidelity (WiFi®) and Bluetooth®), cable TV, satellite links and the like.

[0062] In some implementations, server computers **102** may not be necessary and/or preferred. For example, in one or more implementations, a stand-alone gaming device such as gaming device **104**A, gaming device **104**B or any of the other gaming devices **104**C-**104**X can implement one or more aspects of the present disclosure. However, it is typical to find multiple EGMs connected to networks implemented with one or more of the different server computers **102** described herein.

[0063] The server computers **102** may include a central determination gaming system server **106**, a ticket-in-ticket-out (TITO) system server **108**, a player tracking system server **110**, a progressive system server **112**, and/or a casino management system server **114**. Gaming devices **104**A-**104**X may include features to enable operation of any or all servers for use by the player and/or operator (e.g., the casino, resort, gaming establishment, tavern, pub, etc.). For example, game outcomes may be generated on a central determination gaming system server **106** and then transmitted over the network to any of a group of remote terminals or remote gaming devices **104**A-**104**X that utilize the game outcomes and display the results to the players.

[0064] Gaming device **104**A is often of a cabinet construction which may be aligned in rows or banks of similar devices for placement and operation on a casino floor. The gaming device **104**A often includes a main door which provides access to the interior of the cabinet. Gaming device **104**A typically includes a button area or button deck **120** accessible by a player that is configured with input switches or buttons **122**, an access channel for a bill validator **124**, and/or an access channel for a ticket-out printer **126**.

[0065] In FIG. **1**, gaming device **104**A is shown as a Relm XL™ model gaming device manufactured by Aristocrat® Technologies, Inc. As shown, gaming device **104**A is a reel machine having a gaming display area **118** comprising a number (typically 3 or 5) of mechanical reels **130** with various symbols displayed on them. The mechanical reels **130** are independently spun and stopped to show a set of symbols within the gaming display area **118** which may be used to determine an outcome to the game.

[0066] In many configurations, the gaming device **104**A may have a main display **128** (e.g., video display monitor) mounted to, or above, the gaming display area **118**. The main display **128** can be a high-resolution liquid crystal display (LCD), plasma, light emitting diode (LED), or organic light emitting diode (OLED) panel which may be flat or curved as shown, a cathode ray tube, or other conventional electronically controlled video monitor.

[0067] In some implementations, the bill validator **124** may also function as a “ticket-in” reader that allows the player to use a casino issued credit ticket to load credits onto the gaming device **104**A (e.g., in a cashless ticket (“TITO”) system). In such cashless implementations, the gaming device **104**A may also include a “ticket-out” printer **126** for outputting a credit ticket when a “cash out” button is pressed. Cashless TITO systems are used to generate and track unique bar-codes or other indicators printed on tickets to allow players to avoid the use of bills and coins by loading credits using a ticket reader and cashing out credits using a ticket-out printer **126** on the gaming device **104**A. The gaming device **104**A can have hardware meters for purposes including ensuring regulatory compliance and monitoring the player credit balance. In addition, there can be additional meters that record the total amount of money wagered on the gaming device, total amount of money deposited, total amount of money withdrawn, total amount of winnings on gaming device **104**A.

[0068] In some implementations, a player tracking card reader **144**, a transceiver for wireless communication with a mobile device (e.g., a player's smartphone), a keypad **146**, and/or an illuminated display **148** for reading, receiving, entering, and/or displaying player tracking information is provided in gaming device **104**A. In such implementations, a game controller within the gaming device **104**A can communicate with the player tracking system server **110** to send and receive player tracking information.
[0069] Gaming device **104**A may also include a bonus topper wheel **134**. When bonus play is triggered (e.g., by a player achieving a particular outcome or set of outcomes in the primary game), bonus topper wheel **134** is operative to spin and stop with indicator arrow **136** indicating the outcome of the bonus game. Bonus topper wheel **134** is typically used to play a bonus game, but it could also be incorporated into play of the base or primary game.
[0070] A candle **138** may be mounted on the top of gaming device **104**A and may be activated by a player (e.g., using a switch or one of buttons **122**) to indicate to operations staff that gaming device **104**A has experienced a malfunction or the player requires service. The candle **138** is also often used to indicate a jackpot has been won and to alert staff that a hand payout of an award may be needed.
[0071] There may also be one or more information panels **152** which may be a back-lit, silkscreened glass panel with lettering to indicate general game information including, for example, a game denomination (e.g., $0.25 or $1), pay lines, pay tables, and/or various game related graphics. In some implementations, the information panel(s) **152** may be implemented as an additional video display.
[0072] Gaming devices **104**A have traditionally also included a handle **132** typically mounted to the side of main cabinet **116** which may be used to initiate game play.
[0073] Many or all the above described components can be controlled by circuitry (e.g., a game controller) housed inside the main cabinet **116** of the gaming device **104**A, the details of which are shown in FIG. **2**A.
[0074] An alternative example gaming device **104**B illustrated in FIG. **1** is the Arc™ model gaming device manufactured by Aristocrat® Technologies, Inc. Note that where possible, reference numerals identifying similar features of the gaming device **104**A implementation are also identified in the gaming device **104**B implementation using the same reference numbers. Gaming device **104**B does not include physical reels and instead shows game play functions on main display **128**. An optional topper screen **140** may be used as a secondary game display for bonus play, to show game features or attraction activities while a game is not in play, or any other information or media desired by the game designer or operator. In some implementations, the optional topper screen **140** may also or alternatively be used to display progressive jackpot prizes available to a player during play of gaming device **104**B.
[0075] Example gaming device **104**B includes a main cabinet **116** including a main door which opens to provide access to the interior of the gaming device **104**B. The main or service door is typically used by service personnel to refill the ticket-out printer **126** and collect bills and tickets inserted into the bill validator **124**. The main or service door may also be accessed to reset the machine, verify and/or upgrade the software, and for general maintenance operations.
[0076] Another example gaming device **104**C shown is the Helix™ model gaming device manufactured by Aristocrat® Technologies, Inc. Gaming device **104**C includes a main display **128**A that is in a landscape orientation. Although not illustrated by the front view provided, the main display **128**A may have a curvature radius from top to bottom, or alternatively from side to side. In some implementations, main display **128**A is a flat panel display. Main display **128**A is typically used for primary game play while secondary display **128**B is typically used for bonus game play, to show game features or attraction activities while the game is not in play or any other information or media desired by the game designer or operator. In some implementations, example gaming device **104**C may also include speakers **142** to output various audio such as game sound,

background music, etc.

[0077] Many different types of games, including mechanical slot games, video slot games, video poker, video black jack, video pachinko, keno, bingo, and lottery, may be provided with or implemented within the depicted gaming devices **104**A-**104**C and other similar gaming devices. Each gaming device may also be operable to provide many different games. Games may be differentiated according to themes, sounds, graphics, type of game (e.g., slot game vs. card game vs. game with aspects of skill), denomination, number of paylines, maximum jackpot, progressive or non-progressive, bonus games, and may be deployed for operation in Class 2 or Class 3, etc.

[0078] FIG. **2**A is a block diagram depicting exemplary internal electronic components of a gaming device **200** connected to various external systems. All or parts of the gaming device **200** shown could be used to implement any one of the example gaming devices **104**A-X depicted in FIG. **1**. As shown in FIG. **2**A, gaming device **200** includes a topper display **216** or another form of a top box (e.g., a topper wheel, a topper screen, etc.) that sits above cabinet **218**. Cabinet **218** or topper display **216** may also house a number of other components which may be used to add features to a game being played on gaming device **200**, including speakers **220**, a ticket printer **222** which prints bar-coded tickets or other media or mechanisms for storing or indicating a player's credit value, a ticket reader **224** which reads bar-coded tickets or other media or mechanisms for storing or indicating a player's credit value, and a player tracking interface **232**. Player tracking interface **232** may include a keypad **226** for entering information, a player tracking display **228** for displaying information (e.g., an illuminated or video display), a card reader **230** for receiving data and/or communicating information to and from media or a device such as a smart phone enabling player tracking. FIG. **2** also depicts utilizing a ticket printer **222** to print tickets for a TITO system server **108**. Gaming device **200** may further include a bill validator **234**, player-input buttons **236** for player input, cabinet security sensors **238** to detect unauthorized opening of the cabinet **218**, a primary game display **240**, and a secondary game display **242**, each coupled to and operable under the control of game controller **202**.

[0079] The games available for play on the gaming device **200** are controlled by a game controller **202** that includes one or more processors **204**. Processor **204** represents a general-purpose processor, a specialized processor intended to perform certain functional tasks, or a combination thereof. As an example, processor **204** can be a central processing unit (CPU) that has one or more multi-core processing units and memory mediums (e.g., cache memory) that function as buffers and/or temporary storage for data. Alternatively, processor **204** can be a specialized processor, such as an application specific integrated subsystem (ASIC), graphics processing unit (GPU), field-programmable gate array (FPGA), digital signal processor (DSP), or another type of hardware accelerator. In another example, processor **204** is a system on chip (SoC) that combines and integrates one or more general-purpose processors and/or one or more specialized processors. Although FIG. **2**A illustrates that game controller **202** includes a single processor **204**, game controller **202** is not limited to this representation and instead can include multiple processors **204** (e.g., two or more processors).

[0080] FIG. **2**A illustrates that processor **204** is operatively coupled to memory **208**. Memory **208** is defined herein as including volatile and nonvolatile memory and other types of non-transitory data storage components. Volatile memory is memory that do not retain data values upon loss of power. Nonvolatile memory is memory that do retain data upon a loss of power. Examples of memory **208** include random access memory (RAM), read-only memory (ROM), hard disk drives, solid-state drives, universal serial bus (USB) flash drives, memory cards accessed via a memory card reader, floppy disks accessed via an associated floppy disk drive, optical discs accessed via an optical disc drive, magnetic tapes accessed via an appropriate tape drive, and/or other memory components, or a combination of any two or more of these memory components. In addition, examples of RAM include static random access memory (SRAM), dynamic random access memory (DRAM), magnetic random access memory (MRAM), and other such devices. Examples of ROM include a programmable read-only memory (PROM), an erasable programmable read-only memory (EPROM), an electrically erasable programmable read-only memory (EEPROM), or other like memory device. Even though FIG. **2**A illustrates that game controller **202** includes a single memory **208**, game controller **202** could include multiple memories **208** for storing program instructions and/or data.

[0081] Memory **208** can store one or more game programs **206** that provide program instructions and/or data for carrying out various implementations (e.g., game mechanics) described herein. Stated another way, game program **206** represents an executable program stored in any portion or component of memory **208**. In one or more implementations, game program **206** is embodied in the form of source code that includes human-readable statements written in a programming language or machine code that contains numerical instructions recognizable by a suitable execution system, such as a processor **204** in a game controller or other system. Examples of executable programs include: (1) a compiled program that can be translated into machine code in a format that can be loaded into a random access portion of memory **208** and run by processor **204**; (2) source code that may be expressed in proper format such as object code that is capable of being loaded into a random access portion of memory **208** and executed by processor **204**; and (3) source code that may be interpreted by another executable program to generate instructions in a random access portion of memory **208** to be executed by processor **204**.

[0082] Alternatively, game programs **206** can be set up to generate one or more game instances based on instructions and/or data that gaming device **200** exchanges with one or more remote gaming devices, such as a central determination gaming system server **106** (not shown in FIG. **2**A but shown in FIG. **1**). For purpose of this disclosure, the term "game instance" refers to a play or a round of a game that gaming device **200** presents (e.g., via a user interface (UI)) to a player. The game instance is communicated to gaming device **200** via the network **214** and then displayed on gaming device **200**. For example, gaming device **200** may execute game program **206** as video streaming software that allows the game to be displayed on gaming device **200**. When a game is stored on gaming device **200**, it may be loaded from memory **208** (e.g., from a read only memory (ROM)) or from the central determination gaming system server **106** to memory **208**.

[0083] Gaming devices, such as gaming device **200**, are highly regulated to ensure fairness and, in many cases, gaming device **200** is operable to award monetary awards (e.g., typically dispensed in the form of a redeemable voucher). Therefore, to satisfy security and regulatory requirements in a gaming environment, hardware and software architectures are implemented in gaming devices **200** that differ significantly from those of general-purpose computers. Adapting general purpose computers to function as gaming devices **200** is not simple or straightforward because of: (1) the regulatory requirements for gaming devices **200**, (2) the harsh environment in which gaming devices **200** operate, (3) security requirements, (4) fault tolerance requirements, and (5) the requirement for additional special purpose componentry enabling functionality of an EGM. These differences require substantial engineering effort with respect to game design implementation, game mechanics, hardware components, and software.

[0084] One regulatory requirement for games running on gaming device **200** generally involves complying with a certain level of randomness. Typically, gaming jurisdictions mandate that gaming devices **200** satisfy a minimum level of randomness without specifying how a gaming device **200** should achieve this level of randomness. To comply, FIG. **2**A illustrates that gaming device **200** could include an RNG **212** that utilizes hardware and/or software to generate RNG outcomes that lack any pattern. The RNG operations are often specialized and non-generic in order to comply with regulatory and gaming requirements. For example, in a slot game, game program **206** can initiate multiple RNG calls to RNG **212** to generate RNG outcomes, where each RNG call and RNG outcome corresponds to an outcome for a reel. In another example, gaming device **200** can be a Class II gaming device where RNG **212** generates RNG outcomes for creating Bingo cards. In one or more implementations, RNG **212** could be one of a set of RNGs operating on gaming device

200. More generally, an output of the RNG **212** can be the basis on which game outcomes are determined by the game controller **202**. Game developers could vary the degree of true randomness for each RNG (e.g., pseudorandom) and utilize specific RNGs depending on game requirements. The output of the RNG **212** can include a random number or pseudorandom number (either is generally referred to as a "random number").

[0085] In FIG. **2**A, RNG **212** and hardware RNG **244** are shown in dashed lines to illustrate that RNG **212**, hardware RNG **244**, or both can be included in gaming device **200**. In one implementation, instead of including RNG **212**, gaming device **200** could include a hardware RNG **244** that generates RNG outcomes. Analogous to RNG **212**, hardware RNG **244** performs specialized and non-generic operations in order to comply with regulatory and gaming requirements. For example, because of regulation requirements, hardware RNG **244** could be a random number generator that securely produces random numbers for cryptography use. The gaming device **200** then uses the secure random numbers to generate game outcomes for one or more game features. In another implementation, the gaming device **200** could include both hardware RNG **244** and RNG **212**. RNG **212** may utilize the RNG outcomes from hardware RNG **244** as one of many sources of entropy for generating secure random numbers for the game features.

[0086] Another regulatory requirement for running games on gaming device **200** includes ensuring a certain level of RTP. Similar to the randomness requirement discussed above, numerous gaming jurisdictions also mandate that gaming device **200** provides a minimum level of RTP (e.g., RTP of at least 75%). A game can use one or more lookup tables (also called weighted tables) as part of a technical solution that satisfies regulatory requirements for randomness and RTP. In particular, a lookup table can integrate game features (e.g., trigger events for special modes or bonus games; newly introduced game elements such as extra reels, new symbols, or new cards; stop positions for dynamic game elements such as spinning reels, spinning wheels, or shifting reels; or card selections from a deck) with random numbers generated by one or more RNGs, so as to achieve a given level of volatility for a target level of RTP. (In general, volatility refers to the frequency or probability of an event such as a special mode, payout, etc. For example, for a target level of RTP, a higher-volatility game may have a lower payout most of the time with an occasional bonus having a very high payout, while a lower-volatility game has a steadier payout with more frequent bonuses of smaller amounts.) Configuring a lookup table can involve engineering decisions with respect to how RNG outcomes are mapped to game outcomes for a given game feature, while still satisfying regulatory requirements for RTP. Configuring a lookup table can also involve engineering decisions about whether different game features are combined in a given entry of the lookup table or split between different entries (for the respective game features), while still satisfying regulatory requirements for RTP and allowing for varying levels of game volatility.

[0087] FIG. **2**A illustrates that gaming device **200** includes an RNG conversion engine **210** that translates the RNG outcome from RNG **212** to a game outcome presented to a player. To meet a designated RTP, a game developer can set up the RNG conversion engine **210** to utilize one or more lookup tables to translate the RNG outcome to a symbol element, stop position on a reel strip layout, and/or randomly chosen aspect of a game feature. As an example, the lookup tables can regulate a prize payout amount for each RNG outcome and how often the gaming device **200** pays out the prize payout amounts. The RNG conversion engine **210** could utilize one lookup table to map the RNG outcome to a game outcome displayed to a player and a second lookup table as a pay table for determining the prize payout amount for each game outcome. The mapping between the RNG outcome to the game outcome controls the frequency in hitting certain prize payout amounts.

[0088] FIG. **2**A also depicts that gaming device **200** is connected over network **214** to player tracking system server **110**. Player tracking system server **110** may be, for example, an OASIS® system manufactured by Aristocrat® Technologies, Inc. Player tracking system server **110** is used to track play (e.g., amount wagered, games played, time of play and/or other quantitative or qualitative measures) for individual players so that an operator may reward players in a loyalty program. The player may use the player tracking interface **232** to access his/her account information, activate free play, and/or request various information. Player tracking or loyalty programs seek to reward players for their play and help build brand loyalty to the gaming establishment. The rewards typically correspond to the player's level of patronage (e.g., to the player's playing frequency and/or total amount of game plays at a given casino). Player tracking rewards may be complimentary and/or discounted meals, lodging, entertainment and/or additional play. Player tracking information may be combined with other information that is now readily obtainable by a casino management system.

[0089] When a player wishes to play the gaming device **200**, he/she can insert cash or a ticket voucher through a coin acceptor (not shown) or bill validator **234** to establish a credit balance on the gaming device. The credit balance is used by the player to place wagers on instances of the game and to receive credit awards based on the outcome of winning instances. The credit balance is decreased by the amount of each wager and increased upon a win. The player can add additional credits to the balance at any time. The player may also optionally insert a loyalty club card into the card reader **230**. During the game, the player views with one or more UIs, the game outcome on one or more of the primary game display **240** and secondary game display **242**. Other game and prize information may also be displayed.

[0090] For each game instance, a player may make selections, which may affect play of the game. For example, the player may vary the total amount wagered by selecting the amount bet per line and the number of lines played. In many games, the player is asked to initiate or select options during course of game play (such as spinning a wheel to begin a bonus round or select various items during a feature game). The player may make these selections using the player-input buttons **236**, the primary game display **240** which may be a touch screen, or using some other device which enables a player to input information into the gaming device **200**.

[0091] During certain game events, the gaming device **200** may display visual and auditory effects that can be perceived by the player. These effects add to the excitement of a game, which makes a player more likely to enjoy the playing experience. Auditory effects include various sounds that are projected by the speakers **220**. Visual effects include flashing lights, strobing lights or other patterns displayed from lights on the gaming device **200** or from lights behind the information panel **152** (FIG. **1**).

[0092] When the player is done, he/she cashes out the credit balance (typically by pressing a cash out button to receive a ticket from the ticket printer **222**). The ticket may be “cashed-in” for money or inserted into another machine to establish a credit balance for play.

[0093] Additionally, or alternatively, gaming devices **104**A-**104**X and **200** can include or be coupled to one or more wireless transmitters, receivers, and/or transceivers (not shown in FIGS. **1** and **2**A) that communicate (e.g., Bluetooth® or other near-field communication technology) with one or more mobile devices to perform a variety of wireless operations in a casino environment. Examples of wireless operations in a casino environment include detecting the presence of mobile devices, performing credit, points, comps, or other marketing or hard currency transfers, establishing wagering sessions, and/or providing a personalized casino-based experience using a mobile application. In one implementation, to perform these wireless operations, a wireless transmitter or transceiver initiates a secure wireless connection between a gaming device **104**A-**104**X and **200** and a mobile device. After establishing a secure wireless connection between the gaming device **104**A-**104**X and **200** and the mobile device, the wireless transmitter or transceiver does not send and/or receive application data to and/or from the mobile device. Rather, the mobile device communicates with gaming devices **104**A-**104**X and **200** using another wireless connection (e.g., WiFi® or cellular network). In another implementation, a wireless transceiver establishes a secure connection to directly communicate with the mobile device. The mobile device and gaming device **104**A-**104**X and **200** sends and receives data utilizing the wireless transceiver instead of utilizing an external network. For example, the mobile device would perform digital wallet transactions by directly communicating with the wireless transceiver. In one or more implementations, a wireless transmitter could broadcast data received by one or more mobile devices without establishing a pairing connection with the mobile devices.

[0094] Although FIGS. **1** and **2**A illustrate specific implementations of a gaming device (e.g., gaming devices **104**A-**104**X and **200**), the disclosure is not limited to those implementations shown in FIGS. **1** and **2**. For example, not all gaming devices suitable for implementing implementations of the present disclosure necessarily include top wheels, top boxes, information panels, cashless ticket systems, and/or player tracking systems. Further, some suitable gaming devices have only a single game display that includes only a mechanical set of reels and/or a video display, while others are designed for bar counters or tabletops and have displays that face upwards. Gaming devices **104**A-**104**X and **200** may also include other processors that are not separately shown. Using FIG. **2**A as an example, gaming device **200** could include display controllers (not shown in FIG. **2**A) configured to receive video input signals or instructions to display images on game displays **240** and **242**. Alternatively, such display controllers may be integrated into the game controller **202**. The use and discussion of FIGS. **1** and **2** are examples to facilitate ease of description and explanation.

[0095] FIG. **2**B depicts a casino gaming environment according to one example. In this example, the casino **251** includes banks **252** of EGMs **104**. In this example, each bank **252** of EGMs **104** includes a corresponding gaming signage system **254** (also shown in FIG. **2**A). According to this implementation, the casino **251** also includes mobile gaming devices **256**, which are also configured to present wagering games in this example. The mobile gaming devices **256** may, for example, include tablet devices, cellular phones, smart phones and/or other handheld devices. In this example, the mobile gaming devices **256** are configured for communication with one or more other devices in the casino **251**, including but not limited to one or more of the server computers **102**, via wireless access points **258**.

[0096] According to some examples, the mobile gaming devices **256** may be configured for stand-alone determination of game outcomes. However, in some alternative implementations the mobile gaming devices **256** may be configured to receive game outcomes from another device, such as the central determination gaming system server **106**, one of the EGMs **104**, etc.

[0097] Some mobile gaming devices **256** may be configured to accept monetary credits from a credit or debit card, via a wireless interface (e.g., via a wireless payment app), via tickets, via a patron casino account, etc. However, some mobile gaming devices **256** may not be configured to accept monetary credits via a credit or debit card. Some mobile gaming devices **256** may include a ticket reader and/or a ticket printer whereas some mobile gaming devices **256** may not, depending on the particular implementation.

[0098] In some implementations, the casino **251** may include one or more kiosks **260** that are configured to facilitate monetary transactions involving the mobile gaming devices **256**, which may include cash out and/or cash in transactions. The kiosks **260** may be configured for wired and/or wireless communication with the mobile gaming devices **256**. The kiosks **260** may be configured to accept monetary credits from casino patrons **262** and/or to dispense monetary credits to casino patrons **262** via cash, a credit or debit card, via a wireless interface (e.g., via a wireless payment app), via tickets, etc. According to some examples, the kiosks **260** may be configured to accept monetary credits from a casino patron and to provide a corresponding amount of monetary credits to a mobile gaming device **256** for wagering purposes, e.g., via a wireless link such as a near-field communications link. In some such examples, when a casino patron **262** is ready to cash out, the casino patron **262** may select a cash out option provided by a mobile gaming device **256**, which may include a real button or a virtual button (e.g., a button provided via a graphical user interface) in some instances. In some such examples, the mobile gaming device **256** may send a “cash out” signal to a kiosk **260** via a wireless link in response to receiving a “cash out” indication from a casino patron. The kiosk **260** may provide monetary credits to the casino patron **262** corresponding to the “cash out” signal, which may be in the form of cash, a credit ticket, a credit transmitted to a financial account corresponding to the casino patron, etc.

[0099] In some implementations, a cash-in process and/or a cash-out process may be facilitated by the TITO system server **108**. For example, the TITO system server **108** may control, or at least authorize, ticket-in and ticket-out transactions that involve a mobile gaming device **256** and/or a kiosk **260**.

[0100] Some mobile gaming devices **256** may be configured for receiving and/or transmitting player loyalty information. For example, some mobile gaming devices **256** may be configured for wireless communication with the player tracking system server **110**. Some mobile gaming devices **256** may be configured for receiving and/or transmitting player loyalty information via wireless communication with a patron's player loyalty card, a patron's smartphone, etc.

[0101] According to some implementations, a mobile gaming device **256** may be configured to provide safeguards that prevent the mobile gaming device **256** from being used by an unauthorized person. For example, some mobile gaming devices **256** may include one or more biometric sensors and may be configured to receive input via the biometric sensor(s) to verify the identity of an authorized patron. Some mobile gaming devices **256** may be configured to function only within a predetermined or configurable area, such as a casino gaming area.

[0102] FIG. **2**C is a diagram that shows examples of components of a system for providing online gaming according to some aspects of the present disclosure. As with other figures presented in this disclosure, the numbers, types and arrangements of gaming devices shown in FIG. **2**C are merely shown by way of example. In this example, various gaming devices, including but not limited to end user devices (EUDs) **264***a*, **264***b* and **264***c* are capable of communication via one or more networks **417**. The networks **417** may, for example, include one or more cellular telephone networks, the Internet, etc. In this example, the EUDs **264***a* and **264***b* are mobile devices: according to this example the EUD **264***a* is a tablet device and the EUD **264***b* is a smart phone. In this implementation, the EUD **264***c* is a laptop computer that is located within a residence **266** at the time depicted in FIG. **2**C. Accordingly, in this example the hardware of EUDs is not specifically configured for online gaming, although each EUD is configured with software for online gaming. For example, each EUD may be configured with a web browser. Other implementations may include other types of EUD, some of which may be specifically configured for online gaming.

[0103] In this example, a gaming data center **276** includes various devices that are configured to provide online wagering games via the networks **417**. The gaming data center **276** is capable of communication with the networks **417** via the gateway **272**. In this example, switches **278** and routers **280** are configured to provide network connectivity for devices of the gaming data center **276**, including storage devices **282***a*, servers **284***a* and one or more workstations **570***a*. The servers **284***a* may, for example, be configured to provide access to a library of games for online game play. In some examples, code for executing at least some of the games may initially be stored on one or more of the storage devices **282***a*. The code may be subsequently loaded onto a server **284***a* after selection by a player via an EUD and communication of that selection from the EUD via the networks **417**. The server **284***a* onto which code for the selected game has been loaded may provide the game according to selections made by a player and indicated via the player's EUD. In other examples, code for executing at least some of the games may initially be stored on one or more of the servers **284***a*. Although only one gaming data center **276** is shown in FIG. **2**C, some implementations may include multiple gaming data centers **276**.

[0104] In this example, a financial institution data center **270** is also configured for communication via the networks **417**. Here, the financial institution data center **270** includes servers **284***b*, storage devices **282***b*, and one or more workstations **286***b*. According to this example, the financial institution data center **270** is configured to maintain financial accounts, such as checking accounts, savings accounts, loan accounts, etc. In some implementations one or more of the authorized users **274***a*-**274***c* may maintain at least one financial account with the financial institution that is serviced via the financial institution data center **270**.

[0105] According to some implementations, the gaming data center **276** may be configured to provide online wagering games in which money may be won or lost. According to some such implementations, one or more of the servers **284***a* may be configured to monitor player credit balances, which may be expressed in game credits, in currency units, or in any other appropriate manner. In some implementations, the server(s) **284***a* may be configured to obtain financial credits from and/or provide financial credits to one or more financial institutions, according to a player's “cash in” selections, wagering game results and a player's “cash out” instructions. According to some such implementations, the server(s) **284***a* may be configured to electronically credit or debit the account of a player that is maintained by a financial institution, e.g., an account that is maintained via the financial institution data center **270**. The server(s) **284***a* may, in some examples, be configured to maintain an audit record of such transactions.

[0106] In some alternative implementations, the gaming data center **276** may be configured to provide online wagering games for which credits may not be exchanged for cash or the equivalent. In some such examples, players may purchase game credits for online game play, but may not “cash out” for monetary credit after a gaming session. Moreover, although the financial institution data center **270** and the gaming data center **276** include their own servers and storage devices in this example, in some examples the financial institution data center **270** and/or the gaming data center **276** may use offsite “cloud-based” servers and/or storage devices. In some alternative examples, the financial institution data center **270** and/or the gaming data center **276** may rely entirely on cloud-based servers.

[0107] One or more types of devices in the gaming data center **276** (or elsewhere) may be capable of executing middleware, e.g., for data management and/or device communication. Authentication information, player tracking information, etc., including but not limited to information obtained by EUDs **264** and/or other information regarding authorized users of EUDs **264** (including but not limited to the authorized users **274***a*-**274***c*), may be stored on storage devices **282** and/or servers **284**. Other game-related information and/or software, such as information and/or software relating to leaderboards, players currently playing a game, game themes, game-related promotions, game competitions, etc., also may be stored on storage devices **282** and/or servers **284**. In some implementations, some such game-related software may be available as “apps” and may be downloadable (e.g., from the gaming data center **276**) by authorized users.

[0108] In some examples, authorized users and/or entities (such as representatives of gaming regulatory authorities) may obtain gaming-related information via the gaming data center **276**. One or more other devices (such EUDs **264** or devices of the gaming data center **276**) may act as intermediaries for such data feeds. Such devices may, for example, be capable of applying data filtering algorithms, executing data summary and/or analysis software, etc. In some implementations, data filtering, summary and/or analysis software may be available as “apps” and downloadable by authorized users.

[0109] FIG. **3** illustrates, in block diagram form, an implementation of a game processing architecture **300** that implements a game processing pipeline for the play of a game in accordance with various implementations described herein. As shown in FIG. **3**, the gaming processing pipeline starts with having a UI system **302** receive one or more player inputs for the game instance. Based on the player input(s), the UI system **302** generates and sends one or more RNG calls to a game processing backend system **314**. Game processing backend system **314** then processes the RNG calls with RNG engine **316** to generate one or more RNG outcomes. The RNG outcomes are then sent to the RNG conversion engine **320** to generate one or more game outcomes for the UI system **302** to display to a player. The game processing architecture **300** can implement the game processing pipeline using a gaming device, such as gaming devices **104**A-**104**X and **200** shown in FIGS. **1** and **2**, respectively. Alternatively, portions of the gaming processing architecture **300** can implement the game processing pipeline using a gaming device and one or more remote gaming devices, such as central determination gaming system server **106** shown in FIG. **1**.

[0110] The UI system **302** includes one or more UIs that a player can interact with. The UI system **302** could include one or more game play UIs **304**, one or more bonus game play UIs **308**, and one or more multiplayer UIs **312**, where each UI type includes one or more mechanical UIs and/or graphical UIs (GUIs). In other words, game play UI **304**, bonus game play UI **308**, and the multiplayer UI **312** may utilize a variety of UI elements, such as mechanical UI elements (e.g., physical “spin” button or mechanical reels) and/or GUI elements (e.g., virtual reels shown on a video display or a virtual button deck) to receive player inputs and/or present game play to a player. Using FIG. **3** as an example, the different UI elements are shown as game play UI elements **306**A-**306**N and bonus game play UI elements **310**A-**310**N.

[0111] The game play UI **304** represents a UI that a player typically interfaces with for a base game. During a game instance of a base game, the game play UI elements **306**A-**306**N (e.g., GUI elements depicting one or more virtual reels) are shown and/or made available to a user. In a subsequent game instance, the UI system **302** could transition out of the base game to one or more bonus games. The bonus game play UI **308** represents a UI that utilizes bonus game play UI elements **310**A-**310**N for a player to interact with and/or view during a bonus game. In one or more implementations, at least some of the game play UI element **306**A-**306**N are similar to the bonus game play UI elements **310**A-**310**N. In other implementations, the game play UI element **306**A-**306**N can differ from the bonus game play UI elements **310**A-**310**N.

[0112] FIG. **3** also illustrates that UI system **302** could include a multiplayer UI **312** purposed for game play that differs or is separate from the typical base game. For example, multiplayer UI **312** could be set up to receive player inputs and/or presents game play information relating to a tournament mode. When a gaming device transitions from a primary game mode that presents the base game to a tournament mode, a single gaming device is linked and synchronized to other gaming devices to generate a tournament outcome. For example, multiple RNG engines **316** corresponding to each gaming device could be collectively linked to determine a tournament outcome. To enhance a player's gaming experience, tournament mode can modify and synchronize sound, music, reel spin speed, and/or other operations of the gaming devices according to the tournament game play. After tournament game play ends, operators can switch back the gaming device from tournament mode to a primary game mode to present the base game. Although FIG. **3** does not explicitly depict that multiplayer UI **312** includes UI elements, multiplayer UI **312** could also include one or more multiplayer UI elements.

[0113] Based on the player inputs, the UI system **302** could generate RNG calls to a game processing backend system **314**. As an example, the UI system **302** could use one or more application programming interfaces (APIs) to generate the RNG calls. To process the RNG calls, the RNG engine **316** could utilize gaming RNG **318** and/or non-gaming RNGs **319**A-**319**N. Gaming RNG **318** could corresponds to RNG **212** or hardware RNG **244** shown in FIG. **2**A. As previously discussed with reference to FIG. **2**A, gaming RNG **318** often performs specialized and non-generic operations that comply with regulatory and/or game requirements. For example, because of regulation requirements, gaming RNG **318** could correspond to RNG **212** by being a cryptographic RNG or pseudorandom number generator (PRNG) (e.g., Fortuna PRNG) that securely produces random numbers for one or more game features. To securely generate random numbers, gaming RNG **318** could collect random data from various sources of entropy, such as from an operating system (OS) and/or a hardware RNG (e.g., hardware RNG **244** shown in FIG. **2**A). Alternatively, non-gaming RNGs **319**A-**319**N may not be cryptographically secure and/or be computationally less expensive. Non-gaming RNGs **319**A-**319**N can, thus, be used to generate outcomes for non-gaming purposes. As an example, non-gaming RNGs **319**A-**319**N can generate random numbers for generating random messages that appear on the gaming device.

[0114] The RNG conversion engine **320** processes each RNG outcome from RNG engine **316** and converts the RNG outcome to a UI outcome that is feedback to the UI system **302**. With reference to FIG. **2**A, RNG conversion engine **320** corresponds to RNG conversion engine **210** used for game play. As previously described, RNG conversion engine **320** translates the RNG outcome from the RNG **212** to a game outcome presented to a player. RNG conversion engine **320** utilizes one or more lookup tables **322**A-**322**N to regulate a prize payout amount for each RNG outcome and how often the gaming device pays out the derived prize payout amounts. In one example, the RNG conversion engine **320** could utilize one lookup table to map the RNG outcome to a game outcome displayed to a player and a second lookup table as a pay table for determining the prize payout amount for each game outcome. In this example, the mapping between the RNG outcome and the game outcome controls the frequency in hitting certain prize payout amounts. Different lookup tables could be utilized depending on the different game modes, for example, a base game versus a bonus game.

[0115] After generating the UI outcome, the game processing backend system **314** sends the UI outcome to the UI system **302**. Examples of UI outcomes are symbols to display on a video reel or reel stops for a mechanical reel. In one example, if the UI outcome is for a base game, the UI system **302** updates one or more game play UI elements **306**A-**306**N, such as symbols, for the game play UI **304**. In another example, if the UI outcome is for a bonus game, the UI system could update one or more bonus game play UI elements **310**A-**310**N (e.g., symbols) for the bonus game play UI **308**. In response to updating the appropriate UI, the player may subsequently provide additional player inputs to initiate a subsequent game instance that progresses through the game processing pipeline.

[0116] FIG. **4** illustrates, in block diagram form, an example gaming device **400** in accordance with various implementations described herein. The gaming device **400** has a cabinet construction which may be aligned with other similar gaming devices in rows or banks for placement and operation on a casino floor. As described in detail below, the gaming device **400** includes a host control module **402** (e.g., a gaming control box), a first video display module **404** (e.g., a main video device), and a second video display module **406** (e.g., a top video device) that are connected serially (i.e., daisy-chained). The host control module **402** is communicatively and electrically connected to the first video display module **404** via a first USB interface **408** (e.g., a USB-C cable) for power delivery and data transmission, and the first video display module **404** is communicatively and electrically connected to the second video display module **406** via a second USB interface **410** for power delivery and data transmission. In some implementations, one or both of the USB interfaces **408**, **410** may be USB-C cables configured to carry at least 240 W of power and transfer data at 40 Gbps or higher. The first USB interface **408** and/or the second USB interface **410** may include a cable (e.g., with a USB 2.0 connector, a USB 3.0 connector, a USB 3.1 connector, a USB 3.2 connector, a USB4 connector, a USB4 V2.0 connector, etc.) configured to transmit more or less than 240 W of power and have a data transfer speed that is above or below than 40 Gbps. However, in some implementations, the first USB interface **408** may include a USB 2.0 or higher connector supporting at least 2.5 W of power delivery for powering high-power and high-data devices. As also described in detail below, the gaming device **400** includes one or more non-transitory computer readable media causing components of each module to communicate with one another for implementing power delivery, data transmission, and remedial action when the power required by the components exceeds a maximum power capacity (e.g., 240 W) of either one of the USB-C cables **408**, **410** or a maximum power that the host control module **402** can provide. The use of a single cable for power delivery and data transmission between modules that are connected serially simplifies those connections and thereby simplifies manufacturing, repair or maintenance of the gaming machine, and decreases associated costs.

[0117] The host control module **402** includes a game controller **412** with one or more processors and one or more memory devices. The game controller **412** generates data and/or instructions corresponding with the first video display module **404** and the second video display module **406**. As described in detail below, the game controller **412** further includes a host USB controller **416** having a video logic subsystem communicatively and electrically connected to the game controller **412** and configured to receive data therefrom. The host USB controller **416** further includes an output interface **420** (e.g., USB-C/USB 4.0 interface) for transmitting that data. The host USB controller **416** further includes a power source **414** and a power delivery control subsystem **418** (PD control), which is electrically connected to the power source **414** and configured to receive power therefrom. The host controller **416** delivers power via the output interface **420**. As described in detail below, the host USB controller **416** includes one or more processors and one or more memory devices (e.g., non-transitory computer readable media, etc.) storing instructions which, when executed by one or more processors, cause the one or more processors to cause a first power to be delivered and allocated to components of the first video display module **404** and a second power to be delivered and allocated to components of the second video display module **406**, via the USB-C cables **408**, **410**.

[0118] The first video display module **404** includes a first USB controller **422** communicatively and electrically connected to the host control module **402** via the first USB interface **408**. The first USB interface **408** is configured to deliver a first power (e.g., 240 W) and transmit data from the host USB controller **416** to the first USB controller **422**. The first USB controller **422** has an input interface **424** (e.g., USB-C/USB 4.0 interface) connectable to the output interface **420** of the host USB controller **416** via the first USB interface **408** for power delivery and data transmission between the host control module **402** and the first video display module **404**. The first USB controller **422** further includes a first video logic subsystem **426** communicatively connected to the input interface **424**. The first video logic subsystem **426** is configured to generate an input signal responsive, at least in part to, the first video logic subsystem **426** receiving the data from the host USB controller **416** via the first USB interface **408**. The first video display module **404** further includes a display control subsystem **428** communicatively connected to the first video logic subsystem **426**. The display control subsystem **428** is configured to generate a video signal responsive, at least in part, to the display control subsystem **428** receiving the input signal from the first video logic subsystem **426**. The first video display module **404** further includes a first video monitor **430** communicatively connected to the display control subsystem **428** of the first USB controller **422**. The first video monitor **430** is configured to present videos and/or images corresponding with prize information, gameplay, game outcome, and the like responsive, at least in part, to the first video monitor **430** receiving the video signal from the display control subsystem **428**. In this implementation, the first video monitor **430** has a resolution of at least 3840×2160 at 60 frames per second (i.e., to allows for larger resolution displays, such as 8K). In other implementations, the first video monitor may have other suitable parameters. However, it is contemplated that the monitor may a resolution greater or less than 3840×2160 at 60 frames or more per second (e.g., where the gaming device includes button decks and/or a main monitor is a laser-cut LCD for a custom aspect ratio and thus a custom resolution, etc.).

[0119] In this implementation, one or more first peripheral components **432** are communicatively and electrically connected to the input interface **424** via one or more corresponding ports (e.g., USB 3.0, USB 2.0, etc.) to receive data and at least some of the first power from the host USB controller **416** via the first USB interface **408**. The first peripheral components **432** may be components that are separate from the first video display module **404**. In other implementations, the first peripheral components **432** may be integral parts of the first video display module **404**. One or more examples of the first peripheral components **432** may include a touchscreen display, a lighting device, a microphone, a camera, a speaker, and/or a supplemental monitor. In some implementations, the first peripheral components **432** may include at least one of a DisplayPort for a monitor having a resolution of at least 3840×2160 at 60 frames per second, a USB-C Power Delivery for a possible additional monitor instead of using the second DisplayPort, and/or a power output for the monitor via a Molex/Amphenol header. The first USB controller **422** further includes an output interface **434** (e.g., USB-C/USB 4.0 interface) for transmitting certain data that the input interface **424** receives from the output interface **420** of the host USB controller **416** via the first USB interface **408**. In other implementations, the monitor may have a resolution of less than 3840×2160 at 60 frames or more per second. In yet other implementations, other examples of the first peripheral components **432** may include an audio input to an amplifier via USB or onboard audio input via local PCB traces. In still other implementations, other examples of the first peripheral components **432** may include components of a button deck (e.g., a QI wireless charging device, one or more buttons such as a main bash button, LED lighting on the button deck, a vibration/haptic feedback device, etc.).

[0120] The first video display module **404** further includes a first power delivery control subsystem **436** (first PD control), which is electrically connected to the input interface **424** to receive at least the first power from the host USB controller **416** via the first USB interface **408**. As described in detail below, the first video monitor **430** is electrically connected to the first PD control **436** to receive at least some of the first power (e.g., the first power corresponding with the first video display module **404**) from the host USB controller **416**. The first peripheral components **432** are electrically connected to the first PD control **436** via USB 3.0 or USB 2.0 connection to receive at least some of the first power therefrom.

[0121] In a somewhat analogous manner, the second video display module **406** includes a second USB controller **438** communicatively and electrically connected to the first USB controller **422** of the first video display module **404** via the second USB interface **410**. The second USB interface **410** is configured to transmit data between the first USB controller **422** and the second USB controller **438**. The second USB controller **438** has an input interface **440** (e.g., USB-C/USB 4.0 interface) connectable to the output interface **434** of the first video display module **404** via the second USB interface **410** for data input/output/processing. The second USB controller **438** further includes a second video logic subsystem **442** communicatively connected to the input interface **440**. The second video logic subsystem **442** is configured to generate an input signal responsive, at least in part to, the second video logic subsystem **442** receiving the data from the first USB controller **422** via the second USB interface **410**. The second video display module **406** further includes a display control subsystem **444** communicatively connected to the second video logic subsystem **442**. The display control subsystem **444** is configured to generate a video signal responsive, at least in part, to the display control subsystem **444** receiving the input signal from the second video logic subsystem **442**. The second video display module **406** further includes a second video monitor **446** communicatively connected to the display control subsystem **444** of the second USB controller **438**. The second video monitor **446** is configured to present images corresponding with prize information, gameplay, game outcome, and the like responsive, at least in part, to the second video monitor **446** receiving the video signal from the display control subsystem **444**.

[0122] The second video display module **406** further includes a second power delivery control subsystem **448** (second PD control) electrically connected to the input interface **440** to receive power from the first USB controller **422** via the second USB interface **410**, with the first USB controller **422** having received the second power from host USB controller **416** via the first USB interface **408**. As described in detail below, the second video monitor **446** is electrically connected to the second PD control **448** to receive at least some of the second power therefrom. Similarly, the second peripheral components **450** are electrically connected to the second PD control **448** via USB 3.0 or USB 2.0 connection to receive at least some of the second power therefrom.

[0123] In operation, the components of the gaming device **400** communicate with one another (e.g., USB Power Delivery Specification Revision 3.1 via single USB-C cables) to allocate at least some of the first power to first video monitor **430** and the first peripherals and at least some of the second power to the second video monitor **346** and the second peripherals **450**. The second video monitor **446** and the second peripheral components **450** are configured to negotiate with the second USB controller **438** to request the second power therefrom. The second power is at least a sum total of a plurality of second individual power levels corresponding with the second video monitor **446** and the second peripheral components **450** (e.g., 100 W for the second video monitor **446**, 50 W for LEDs, 60 W for speakers, etc.). The second PD control **448** is configured to negotiate, via the corresponding second USB interface **410**, with the first USB controller **422** to request the second power originating from the host USB controller **416**.

[0124] The first video monitor **430** and the first peripheral components **432** are configured to negotiate with the first USB controller **422** to request at least the first power therefrom. The first power is the sum total of a plurality of first individual power levels corresponding with the first video monitor **430** and the first peripheral components **432** (e.g., 140 W for the first video monitor **430**, 60 W for speakers, etc.). In this implementation, where the host control module **402**, the first video display module **404**, and the second video display module **406** are connected in a serial combination, the first PD control **436** is configured to negotiate, via the first USB interface **408**, with the host USB controller **416** to request the sum total of the first power and the second power from the host USB controller **416**.

[0125] In some implementations, the host USB controller **416** includes the memory devices that store computer-executable instructions which, when executed by the one or more processors, further cause the one or more processors to cause the host USB controller **416** to compare the sum total of the first power and the second power to a maximum power level capacity of the first USB interface **408** and/or the second USB interface **410** (e.g., 240 W) responsive, at least in part, to the host USB controller **416** receiving the request for the amount of power (e.g., the sum total of the first power and the second power). The memory devices store additional computer-executable instructions which, when executed by the one or more processors, further cause the one or more processors to cause the host USB controller **416** to deliver at least the first power (e.g., the first power and the second power) to the first USB controller **422** responsive, at least in part, to the host USB controller **416** determining that the requested sum total of power is less than the maximum power capacity of the USB interface **408** and/or the USB interface **410**. The memory devices store additional computer-executable instructions which, when executed by the one or more processors, further cause the one or more processors to cause the first USB controller **422** to deliver at least the second power to the second USB controller **438** responsive, at least in part, to the host USB controller **416** determining that the requested sum total of the first power and the second power is less than the maximum power capacity of the USB interface **408** and/or the USB interface **410**.

[0126] The memory devices store additional computer-executable instructions which, when executed by the one or more processors, further cause the host USB controller **416** to generate data associated with a remedial action responsive, at least in part, to the host USB controller **416** determining that the requested sum total of the first power and the second power exceeds the maximum power level capacity of the first USB interface **408** and/or the second USB interface **410**. The first video logic subsystem **426** of the first USB controller **422** is configured to generate an input signal responsive, at least in part, to the first video logic subsystem receiving data associated with the remedial action from the host USB controller **416**. The display control subsystem **428** is configured to generate a video signal responsive, at least in part, to the display control subsystem **428** receiving that input signal from the first video logic subsystem **426**. The first video monitor **430** presents a notification responsive, at least in part, to the first video monitor **430** receiving the video signal from the display control subsystem **428**. The notification prompts an installation of an additional USB-C cable to connect the host USB controller **416** with the first video display module **404** or the second video display module **406**. In some implementations, the first PD control **436** may be configured to dynamically allocate power to the components of the gaming device **400** (e.g., by stopping delivery of power to one or more of the first peripheral components **432**, etc.) responsive, at least in part, to the first USB controller **422** receiving the data associated with the remedial action from the host USB controller **416**. In other implementations, the remedial action may include causing the first USB controller **422** and/or the second USB controller **438** to transmit an over-power signal to, for example, the game controller **412**. The over-power signal may identify one or more peripheral components **432**, **450** as being the source of the fault (e.g., by identifying one or more peripheral components **432**, **450** that cause the total requested power to exceed the maximum power capacity of the first USB interface **408** and/or the second USB interface **410**).

[0127] In another somewhat analogous implementation, the second video logic subsystem **442** is configured to generate an input signal responsive, at least in part, to the second video logic subsystem **422** receiving the data associated with the remedial action from the first USB controller **422**. The display control subsystem **444** of the second USB controller **438** is configured to generate a video signal responsive, at least in part, to the display control subsystem **444** receiving that input signal from the second video logic subsystem **442**. The second video monitor **446** presents a notification responsive, at least in part, to the second video monitor **446** receiving the video signal from the display control subsystem **444**. The notification prompts an installation of an additional USB-C cable to connect the host USB controller **416** with the first video display module **404** or the second video display module **406**. In some implementations, the second PD control **448** may be configured to dynamically allocate power to the components of the gaming device **400** (e.g., by stopping delivery of power to one or more of the second peripheral components **450**, etc.) responsive, at least in part, to the second USB controller **438** receiving the data associated with the remedial action from the first USB controller **422**.

[0128] The gaming device **400** includes a Unified USB Data and Power Distribution System having the host USB controller **416**, the first USB controller **422**, and the second USB controller **438**. The Unified USB Data and Power Distribution includes the plurality of processors and the plurality of memory devices (e.g., non-transitory computer readable media or CRM). The one or more memory devices store computer-executable instructions which, when executed by the one or more processors, cause the components as described above to communicate with one another for power delivery and data transmission. More specifically, the one or more memory devices store computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the second video monitor **446** and the one or more second peripheral components to negotiate with the second USB controller **438** to request at least the second power originating from the host USB controller **408**. The second power is at least the total sum of second individual power levels corresponding with the second video monitor **446** and the one or more second peripheral components **450**. The one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the second power delivery subsystem **448** of the second USB controller **438** to negotiate, via the corresponding second USB interface **410**, with the first USB controller **422** to request the second power (e.g., 100 W for the second video monitor **446**).

[0129] The one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the first video monitor **430** and the one or more first peripheral components **432** to negotiate with the first USB controller **422** to request the first power originating from the host USB controller **416**. The first power is at least a sum total of first individual power levels corresponding with the first video monitor **430** and the one or more first peripheral components **432**. The one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the first power delivery subsystem **436** of the first USB controller **422** to negotiate, via the first USB interface **408**, with the host USB controller **416** to request the sum total of the first power and the second power (e.g., 140 W for the first video monitor **430** and 100 W for the second video monitor **446**).

[0130] The one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause at least the first power (e.g., the sum of the first power and the second power, etc.) to be delivered, via the first USB interface **408**, from the host USB controller **416** of the host control module **402** to the first USB controller **422** of the first video display module **404** responsive, at least in part, to the

processor receiving from the first USB controller **422** the request for the first power and the second power. The one or more memory devices store computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the first power to be allocated from the first USB controller **422** to the components of the first video display module **404** (e.g., the first video monitor **430**) and the first peripheral components **432** responsive, at least in part, to the first USB controller **422** receiving the amount of power from the host USB controller **416**. The one or more memory devices store computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the second power to be delivered, via the additional second USB interface **410**, from the first USB controller **422** to the second video display module **406** responsive, at least in part, to the first USB controller **422** receiving the amount of power from the host USB controller **416**. The one or more memory devices store computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the second power to be allocated from the second USB controller **438** to the components of the second video display module **406** (e.g., the second video monitor **446** and the second peripheral components **450**) responsive, at least in part, to the second USB controller **438** receiving the amount of power from the first USB controller **422**.
[0131] In other implementations, the one or more memory devices store computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to initiate a remedial action (e.g., dynamically allocate or scale back power delivered to one or more components, provide a notification that prompts an installation of an additional USB-C cable, etc.) when the power level required by the first video display module **404** and the second video display module **406** exceed the maximum power level capacity of either one of the USB-C cables **408**, **410**. The one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause the host USB controller **416** to compare the amount of power (e.g., 714 W for two video monitors, two speakers, and LEDs, etc.) to the maximum power level capacity (e.g., 240 W) of the USB interface **408**. The one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause, responsive, at least in part, to the host USB controller **416** determining that the amount of power exceeds the maximum power level capacity, the host USB controller **416** to generate data associated with a remedial action.
[0132] The one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause, responsive, at least in part, to the first video logic subsystem **426** of the first USB controller **422** receiving the data associated with the remedial action from the host USB controller **416**, the first video logic subsystem **426** to generate an input signal. The one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause, responsive, at least in part, to the display control subsystem **428** of the first video display module **404** receiving the input signal from the first video logic subsystem **426**, the display control subsystem **428** to generate a video signal. The one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause, responsive, at least in part, to the first video monitor **430** receiving the video signal from the display control subsystem **428**, the first video monitor **430** to present a notification. In some implementations, the notification includes a prompt to install an additional USB-C cable to connect the host USB controller **416** with the first video display module **404** or the second video display module **406**. In other implementations, the one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause, responsive, at least in part, to the first power delivery subsystem **436** of the first USB controller **422** receiving the data associated with the remedial action from the host USB controller **416**, the first power delivery

subsystem **436** to stop delivering power to one or more of the first peripheral components **432**.
[0133] In a somewhat analogous manner, the one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause, responsive, at least in part, to the second video logic subsystem **442** of the second USB controller **438** receiving the data associated with the remedial action from the first USB controller **422**, the second video logic subsystem **442** to generate an input signal. The one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause, responsive, at least in part, to the display control subsystem **444** of the second video display module **406** receiving the input signal from the second video logic subsystem **442**, the display control subsystem **444** to generate a video signal. The one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause, responsive, at least in part, to the second video monitor **446** receiving the video signal from the display control subsystem **444**, the second video monitor **446** to present a notification. In one implementation, the notification includes a prompt to install an additional USB-C cable to connect the host USB controller **416** with the first video display module **404** or the second video display module **406**. In other implementations, the one or more memory devices store additional computer-executable instructions which, when executed by the one or more processors, cause the one or more processors to cause, responsive, at least in part, to the second power delivery subsystem **448** of the second USB controller **438** receiving the data associated with the remedial action from the first USB controller **422**, the second power delivery subsystem **448** to stop delivering power to one or more of the second peripheral components **450**.
[0134] Referring to FIG. **5**, another example of gaming device **500** is somewhat similar to the gaming device of FIG. **4**. To avoid undue repetition, elements in the implementation of FIG. **5** that are analogous to elements shown in FIG. **4** are called out with numbers that share the same last two digits as those analogous elements in FIG. **4**. Thus, the discussion provided above with respect to the elements of the implementation of FIG. **4** will be understood to be equally applicable to the analogous elements in FIG. **5** unless indicated otherwise. In the interest of conciseness, discussion of these elements that would be redundant of earlier discussion herein of similar elements is not provided, with the understanding that the earlier discussion of such elements is applicable to these similar elements in FIG. **5**.
[0135] While the gaming machine **400** of FIG. **4** includes the host control module **402**, the first video display module **404**, and the second video display module **406** communicatively and electrically connected in serial combination, the gaming machine **500** of FIG. **5** includes the first video display module **504** and the second video display module **506** communicatively and electrically connected directly to the host control module in parallel combination. The second USB controller **406** of FIG. **4** is indirectly connected to the host USB controller **416** via the first USB controller **422**. While the first USB interface **408** of FIG. **4** must carry the total sum of the first power and the second power, the first USB interface **508** of FIG. **5** may carry only the first power corresponding with the first video display module **404**. For that reason, in some implementations, the parallel combination or arrangement of the modules in FIG. **5** may be an additional remedial action when the processors determine that the sum total of power exceeds the maximum power capacity of the first USB interface **508**. Put another way, the first video monitor **530** and/or the second video monitor **546** may present the notification to prompt connecting, using the second USB interface **410**, the second USB controller **438** directly to the host USB controller **416** instead of indirectly connecting the USB controller **538** to the host USB controller via the first USB controller **422** and the first USB interface **408**.
[0136] Referring to FIG. **6**, another example of a gaming machine **600** is somewhat similar to the gaming device **400** of FIG. **4**. To avoid undue repetition, elements in the implementation of FIG. **6** that are analogous to elements shown in FIG. **4** are called out with numbers that share the same last

two digits as those analogous elements in FIG. **4**. Thus, the discussion provided above with respect to the elements of the implementation of FIG. **4** will be understood to be equally applicable to the analogous elements in FIG. **6** unless indicated otherwise. In the interest of conciseness, discussion of these elements that would be redundant of earlier discussion herein of similar elements is not provided, with the understanding that the earlier discussion of such elements is applicable to these similar elements in FIG. **6**.

[0137] While the gaming device **400** of FIG. **4** includes both the first video display module **404** and the second video display module **406**, the gaming device **600** of FIG. **6** includes a single video display module **604** without any additional video display modules. In other implementations, the gaming device **600** may include three or more video display modules.

[0138] FIG. **7** illustrates a flow chart of one example of a method **700** for configuring the gaming device **400** of FIG. **4** to deliver power and data via the USB-C cables **408**, **410**. The method includes using a first USB-C cable **408** to negotiate for power (e.g., maximum power that a host can provide) from the host to a connected first video display module and providing power to a second video display module **406** from the first video display module **404** when the second USB-C cable **410** is connected via negotiation. The maximum power from the host will not be consumed when the second video display module is not connected, even though the first video display module has the capability to deliver that power. The method **700** begins at block **702** with connecting, using the first USB interface **408**, the host control module **402** (e.g., the output interface **420**) to the first video display module **404** (e.g., the input interface **424**). The method **700** then proceeds to block **704**.

[0139] At block **704**, the method **700** includes connecting, using the second USB interface **410**, the first video display module **404** (i.e., via the output interface **434**) to the second video display module **406** (i.e., via the input interface **440**). The method **700** further includes transmitting power from the power source **414** of the host control module to the host USB controller **416** (e.g., the PD control **418**). The method **700** further includes transmitting data from the game controller **412** of the host control module **402** to the host USB controller **416**. The method **700** then proceeds to block **706** to determine power needs for the components of the gaming device **400**.

[0140] At block **706**, the method **700** includes negotiating, using the second PD control **448** of the second USB controller **438**, with the first USB controller **422** via the corresponding second USB interface **410** to request the second power. More specifically, the method **700** includes negotiating, using the second video monitor **446** and the second peripheral components **450** collectively, with the second USB controller **438** to request the second power. The second power is the sum total of the individual power levels corresponding with the second video monitor **446** and/or the second peripheral components **450** (e.g., 100 W for the second video monitor **446**, 50 W for LEDs, etc.). The method **700** then proceeds to block **708**.

[0141] At block **708**, the method **700** includes negotiating, using the first PD control **436** of the first USB controller **422**, with the host USB controller **416** via the first USB interface **408** to request at least the first power (e.g., the sum total of the first power and the second power). More specifically, the sum total of power includes the first power corresponding with the first video display module **404** and the second power corresponding with the second video display module **406**. To identify the sum total of power, the method **700** includes negotiating, using the first video monitor **430** and the first peripheral components **432** collectively, with the first USB controller **422** to request the first power corresponding with the first video monitor **430** and/or the first peripheral components **432** (e.g., 140 W for the second video monitor **446**, 50 W for LEDs, 60 W for speakers, etc.). The first requested power level is at least the sum total of the first individual power levels corresponding with the first video monitor **430** and/or the first peripheral components **432**. The method **700** then proceeds to block **710**.

[0142] At block **710**, the method **700** initiates a remedial action subroutine that includes comparing, using the host USB controller **416**, the amount of power to the maximum power level capacity of the USB interface **408** (e.g., 240 W). In other implementations, the gaming device **400** may use other cables with maximum power level capacities above or below 240 W, and the host USB controller **416** may compare the amount of power with other thresholds above or below the maximum capacity of the USB-C cables **408**, **410** or other cables. If the host USB controller **416** determines that the amount of power does not exceed the maximum power level capacity of the USB interface **408**, the method **700** proceeds to block **712**. If the host USB controller **416** determines that the amount of power exceeds the maximum power level capacity of the USB interface **408**, the method **700** proceeds to block **718**.

[0143] At block **712**, the method **700** includes adjusting, using the host USB controller **416** of the host control module **402**, a voltage level to deliver via the first USB interface **408** the first power to the first USB controller **422** of the first video display module **404**. In one implementation where the USB-C cable implements the USB Power Delivery Specification Revision 3.1, the PD control **418** may adjust the voltage level to 28 Volts to provide a total requested power level of 140 Volts, 36 Volts to provide a total requested power level of 180 W, and **48** Volts to provide a total requested power level of 240 W. The method **700** includes transmitting, using the first USB interface **408**, both the power and the data from the host USB controller **416** to the first USB controller **422** of the first video display module **404**. The method **700** then proceeds to block **714**.

[0144] At block **714**, the method includes transmitting both the power and the data from the first USB controller **422** to the first video monitor **430** and one or more first peripheral components **432** of the first video display module **404**. The first USB controller **422** allocates the first power such that the first video monitor **430** and the first peripheral components **432** receive at least the first power (e.g., the sum total of the individual power levels that were negotiated in block **708**). The method **700** includes further allocating the first power to the first video monitor **430** and the first peripheral components **432** and the second power to the second USB controller **438**. The method **700** includes transmitting, using the corresponding second USB interface **410**, both the power (e.g., the second power level) and the data from the first USB controller **422** to the second USB controller **438** of the second video display module **406**. The method then proceeds to block **716**.

[0145] At block **716**, the method **700** includes transmitting the data and allocating at least some of the second power originating from the host USB controller **416** to the second video monitor **446** and one or more second peripheral components **450** of the second video display module **406**. The second USB controller **438** allocates at least some of the second power to the second video monitor **446** and/or at least some of the second power to the second peripheral components **450** based on the individual power levels that were negotiated in block **706**. The method **700** then returns to block **706**.

[0146] At block **718**, the method **700** includes implementing a remedial action responsive, at least in part, to the host USB controller **416** determining that the amount of power requested by the first video display module **404** and the second video display module **406** (i.e., the sum total of the first power and the second power) exceeds the maximum power level capacity of the USB interface **408**. The method **700** includes generating, using the host USB controller **416**, data associated with a remedial action responsive, at least in part, to the host USB controller **416** determining that the amount of power exceeds the maximum power level capacity. The method **700** further includes generating, using the first video logic subsystem **426** of the first USB controller **422**, the input signal responsive, at least in part, to the first video logic subsystem **426** receiving the data associated with the remedial action from the host USB controller **416**. The method **700** further includes generating, using the display control subsystem **428** of the first video display module **404**, the video signal responsive, at least in part, to the display control subsystem **428** receiving the input signal from the first video logic subsystem **426**. The method **700** further includes presenting, using the first video monitor **430**, the notification responsive, at least in part, to the first video monitor **430** receiving the video signal from the display control subsystem **428**. The method **700** further includes requesting, using the notification, that a technician install an additional USB-C cable to connect the host USB controller **416** with the first video display module **404** or the second video display module **406**.

[0147] In some implementations, the method **700** further includes generating, using the second video logic subsystem **442** of the second USB controller **438**, the input signal responsive, at least in part, to the second video logic subsystem **442** receiving the data associated with the remedial action from the first USB controller **422**. The method **700** further includes generating, using the display control subsystem **444** of the second video display module **406**, the video signal responsive, at least in part, to the display control subsystem **444** receiving the input signal from the second video logic subsystem **442**. The method **700** further includes presenting, using the second video monitor **446**, the notification responsive, at least in part, to the second video monitor **446** receiving the video signal from the display control subsystem **444**. The method **700** further includes prompting, using the notification, an installation of an additional USB-C cable to connect the host USB controller **416** with the first video display module **404** or the second video display module **406**.

[0148] In other implementations, the method **700** further includes stopping, using the first PD control **436** of the first USB controller **422**, delivery of power to the first video monitor and/or one or more of the first peripheral components **432** responsive, at least in part, to the first USB controller **422** receiving the data associated with the remedial action from the host USB controller **416**. The method **700** further includes stopping, using the second PD control **448** of the second USB controller **438**, delivery of power to the second video monitor **446** and/or one or more of the second peripheral components **450** responsive, at least in part, to the second USB controller receiving the data associated with the remedial action from the first USB controller **422**.

[0149] In an embodiment, the USB-C cable may provide greater than 240 W of power to the gaming system **400**. For example, the interface(s) may be operable to convey power at a range of values up to 240 W, although more or less power is considered. Moreover, in some examples, power conversion circuitry may be included in one or more of the interfaces to receive and condition power for use with the associated gaming device, and/or for transmission to a connected interface. In some examples, the host control module **402** may be a computer with a central controller (e.g., processor) connected to and/or controlling operation of the interfaces **420**, **424**, **434**, **440**.

[0150] It is to be understood that the phrases "for each <item>of the one or more <items>," "each <item>of the one or more <items>," or the like, if used herein, are inclusive of both a single-item group and multiple-item groups, i.e., the phrase "for . . . each" is used in the sense that it is used in programming languages to refer to each item of whatever population of items is referenced. For example, if the population of items referenced is a single item, then "each" would refer to only that single item (despite the fact that dictionary definitions of "each" frequently define the term to refer to "every one of two or more things") and would not imply that there must be at least two of those items. Similarly, the term "set" or "subset" should not be viewed, in itself, as necessarily encompassing a plurality of items—it will be understood that a set or a subset can encompass only one member or multiple members (unless the context indicates otherwise).

[0151] In recognition of the possibility of such distributed processing arrangements, the term "collectively," as used herein with reference to memory devices and/or processors or various other items, should be understood to indicate that the referenced collection of items has the characteristics or provides the functionalities that are associated with that collection. For example, if a host USB controller, first USB controller, and second USB controller collectively store instructions for causing A, B, and C to occur, this encompasses at least the following scenarios:

[0152] a) The host USB controller stores instructions for causing A, B, and C to occur, but the first USB controller and/or the second USB controller store no instructions that cause A, B, and C to occur. [0153] b) The first USB controller and/or the second USB controller store instructions for causing A, B, and C to occur, but the host USB controller stores no instructions that cause A, B, and C to occur. [0154] c) The host USB controller stores instructions for causing a proper subset of

A, B, and C to occur, e.g., A and B but not C, and the first USB controller and/or the second USB controller store instructions that cause a different proper subset of A, B, and C to occur, e.g., C but not A and B, where instructions for causing each of A, B, and C to occur are respectively stored on one or more of the host USB controller, the first USB controller, and the second USB controller.
[0155] d) The host USB controller stores instructions for causing a subset of A, B, and C to occur, e.g., A and B but not C, and the first USB controller and/or the second USB controller store instructions that cause a different subset of A, B, and C to occur, e.g., B and C but not A, where instructions for causing each of A, B, and C to occur are respectively stored on one or more of the host USB controller, the first USB controller, and the second USB controller. [0156] e) The host USB controller stores instructions for causing A and a portion of B to occur, and the first USB controller and/or the second USB controller store instructions that cause C and the remaining portion of B to occur.
[0157] In all of the above scenarios, among the host USB controller, the first USB controller, and the second USB controller, there are, collectively, instructions that are stored for causing A, B, and C to occur, i.e., such instructions are stored on one or more USB controllers and it will be recognized that using the term "collectively," e.g., the host USB controller, the first USB controller, and the second USB controller, collectively, store instructions for causing A, B, and C to occur, encompasses all of the above scenarios as well as additional, similar scenarios.
[0158] Similarly, a collection of processors, e.g., a first set of one or more processors and a second set of one or more processors, may be caused, collectively, to perform one or more actions, e.g., actions A, B, and C. As with the previous example, various permutations fall within the scope of such "collective" language: [0159] a) The first set of one or more processors may be caused to perform each of A, B, and C, and the second set of one or more processors may not perform any of A, B, or C. [0160] b) The second set of one or more processors may be caused to perform each of A, B, and C, and the first set of one or more processors may not perform any of A, B, or C. [0161] c) The first set of one or more processors may be caused to perform a proper subset of A, B, and C, and the second set of one or more processors may be caused to perform a different proper subset of A, B, and C to be performed such that between the two sets of processors, all of A, B, and C are caused to be performed. [0162] d) The first set of one or more processors may be caused to perform A and a portion of B, and the second set of one or more processors may be caused to perform C and the remainder of B.
[0163] While the disclosure has been described with respect to the figures, it will be appreciated that many modifications and changes may be made by those skilled in the art without departing from the spirit of the disclosure. Any variation and derivation from the above description and figures are included in the scope of the present disclosure as defined by the claims.

Claims

1. A gaming device comprising: a host control module having a host controller, a game controller communicatively connected to the host controller to transmit data, and a power source electrically connected to the host controller to deliver power; a first video display module having a first controller and a first video monitor communicatively and electrically connected to the first controller; and a second video display module having a second controller and a second video monitor communicatively and electrically connected to the second controller, wherein: the first controller is configured to: receive first power, second power, first data, and second data originating from the host controller, to transmit the first data and deliver at least some of the first power to the first video monitor, and transmit second power and second data to the second controller of the second video display module, and the second controller is configured to: receive second power and second data from the first controller, transmit the second data to the second video monitor, and transmit at least some of the second power received from the first controller to the second video

monitor.
2. The gaming device of claim 1, wherein the host controller is configured to transmit second power and second data to the first controller.
3. The gaming device of claim 1, wherein: the first controller has a first interface configured to transmit the second power and the second data to the second controller, and the second controller has a second interface configured to receive the second power and the second data from the first controller.
4. The gaming device of claim 3, wherein: the first interface is a first USB interface, and the second interface is a second USB interface.
5. The gaming device of claim 4, wherein a USB cable spans between the first USB interface and the second USB interface and is configured to transmit the second power and the second data to the second controller.
6. The gaming device of claim 1, wherein: the host controller has a host interface configured to transmit the first power and the first data to the first controller, and the first controller has a third interface configured to receive the first power, the second power, the first data, and the second data from the host controller.
7. The gaming device of claim 6, wherein the host interface is a host USB interface and the third interface is a third USB interface.
8. The gaming device of claim 1, wherein the first controller is a first USB controller and the second controller is a second USB controller.
9. The gaming device of claim 8, wherein the host controller is a host USB controller.
10. The gaming device of claim 1, wherein the second controller is configured to transmit at least some of the received second power from the first controller to one or more second peripheral components electrically connected to the second controller.
11. The gaming device of claim 10, wherein the second controller has a fourth USB interface electrically connected to the one or more second peripheral components and configured to transmit at least some of the received second power to one or more second peripheral components.
12. A method of configuring a gaming device, the method comprising: transmitting, using a host interface, first power, second power, first data, and second data originating from a host controller to a first interface and a first controller of a first video display module; transmitting, by the first video display module, the received first data and at least some of the received first power to a first video monitor of the first video display module; transmitting, using a second interface of the first video display module, second power and second data to a second controller of a second video display module; and transmitting, by the second video display module, the second data and at least some of the second power received by the second controller to a second video monitor of the second video display module.
13. The method of claim 12, further comprising transmitting, using the second video display module, at least some of the received second power to one or more second peripheral components electrically connected to the second video display module.
14. The method of claim 12, further comprising transmitting, using the first video display module, at least some of the received first power to one or more first peripheral components electrically connected to the first video display module.
15. The method of claim 12, further comprising negotiating, by the second controller, with the first controller to request the second power.
16. The method of claim 12, further comprising negotiating, by the first controller, with the host controller to request the first power and the second power.
17. The method of claim 12, further comprising negotiating, by the second video monitor, with the second controller to request the second power from the first controller.
18. One or more non-transitory computer-readable media storing computer-executable instructions which, when executed by one or more processors, cause the one or more processors to: cause a first

power and a second power originating from a host controller of a host control module to be delivered, via a host interface, to a first controller of a first video display module responsive, at least in part, to the processor receiving from the first controller a request for the first power, cause at least some of the first power to be allocated from the first controller of the first video display module to a first video monitor of the first video display module responsive, at least in part, to the first controller receiving the first power originating from the host controller, cause the second power to be delivered, via a first interface of the first video display module, to a second video display module responsive, at least in part, to the processor receiving from the first controller a request for the second power, and cause at least some of the second power to be allocated from a second controller of the second video display module to a second video monitor of the second video display module responsive, at least in part, to the second controller receiving the second power from the first controller.

19. The one or more non-transitory computer-readable media of claim 18, wherein the one or more computer-readable media further store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to: cause at least some of the second power to be delivered to one or more second peripheral components electrically connected to the second video display module.

20. The one or more non-transitory computer-readable media of claim 19, wherein the one or more computer-readable media further store additional computer-executable instructions which, when executed by one or more processors, cause the one or more processors to: cause the second video monitor and one or more second peripheral components to negotiate with the second controller to collectively request the second power, and the second power being at least a sum total of a plurality of second individual power levels corresponding with the second video monitor and the one or more second peripheral components.